

Auteur: Rob Willemsen
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 12.9

$$\begin{aligned}
 & \int \frac{2^{x+2} + 5^{x-2}}{4^{x-1}} dx \\
 &= \int \frac{2^{x+2}}{4^{x-1}} dx + \int \frac{5^{x-2}}{4^{x-1}} dx \\
 &= \int \frac{2^{x+2}}{2^{2x-2}} dx + \int \frac{\frac{1}{25} \cdot 5^x}{\frac{1}{4} \cdot 4^x} dx \\
 &= \int 2^{4-x} dx + \int \frac{4}{25} \cdot \left(\frac{5}{4}\right)^x dx
 \end{aligned}$$

$$\text{stel } t = 4 - x \Rightarrow dx = -dt$$

$$\begin{aligned}
 &= \int 2^t \cdot (-1) dt + \int \frac{4}{25} \cdot \left(\frac{5}{4}\right)^x dx \\
 &= - \int 2^t dt + \int \frac{4}{25} \cdot \left(\frac{5}{4}\right)^x dx \\
 &= -\frac{2^t}{\ln 2} + \frac{4}{25} \cdot \frac{\left(\frac{5}{4}\right)^x}{\ln\left(\frac{5}{4}\right)} \\
 &= -\frac{2^{4-x}}{\ln 2} + \frac{4}{25} \cdot \frac{\left(\frac{5}{4}\right)^x}{\ln\left(\frac{5}{4}\right)} + C
 \end{aligned}$$

References